

Prediction analysis of Olympic medal table based on PAH comprehensive evaluation model

Summary

This study focuses on the prediction of the Olympic medal table and the analysis of related factors, and is committed to building an effective model to provide guidance for the development of sports in various countries and help them prepare for the 2028 Los Angeles Olympics.

We innovatively constructed the PAH comprehensive evaluation model and ranking-probability model. The PAH comprehensive evaluation model considers the three dimensions of project strength advantage, athlete advantage and home advantage, and calculates the comprehensive evaluation score. The hierarchical analysis method is used to determine the weight of different factors in the gold and medal competition. Subsequently, multivariate linear regression is used to fit the relationship between the comprehensive scores of various projects in previous years and the final medal ownership to predict the 2028 medal table.

The ranking-probability model focuses on the effect of excellent coaches. Based on statistical learning of a large amount of Olympic data, a ranking-probability table is constructed to quantify the contribution of excellent coaches to the team. The linear regression method is used to screen out projects that need to hire excellent coaches for China, the United States, Australia and other countries.

The model system of this study has significant advantages. It has strong scalability and can be widely used in medal prediction, coach selection and sports strategic planning. It has low data dependence and only requires regular Olympic data. It is convenient to adjust parameters and can flexibly adapt to different research needs. Future research can be deepened from multiple directions. Explore nonlinear technology in model fitting; use multi-model methods and optimize weights in prediction strategies; include more international competition results in data.

In summary, this study has made important progress in the field of Olympic medal prediction and related factor analysis, providing valuable references and ideas for subsequent research and providing stronger support and guidance for the development of sports in various countries.

Keywords: Olympic medal prediction, PAH evaluation model, ranking-probability model, multivariate linear regression, sports decision-making

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1 Introduction

1.1 Problem Background

When the flame of the 2024 Paris Olympics slowly extinguished, the passion for sports did not dissipate. The world's attention has been focused on the next place where the light of the Olympics shines - Los Angeles in 2028.



Figure 1 2028 Los Angeles Olympics

In this globally watched sports event, the medal table is undoubtedly one of the focuses of people's attention. We hope to make a reasonable and inspiring prediction of the medal table, and focus on using the model we have constructed to provide guidance and contributions to sports departments and athletes of various countries to better prepare for the 2028 Los Angeles Olympics.

Taking into account the background information and the constraints specified in the problem statement, we are required to address the following issues:

1. Develop a model that can predict medal counts for each country, estimate the model's accuracy, and provide indicators of the model's performance.
2. Consider the impact of good coaches on national performance in this event. Provide guidance for the three countries on hiring good coaches and estimate the impact of good coaches.
3. Identify other original insights from the model about Olympic medal ceremonies and explain how this insight can guide local Olympic committees in preparing for the 2028 Los Angeles Olympics.

1.2 Literature Review

In the prediction of Olympic medals, previous work was mainly carried out according to the following methods:

- Time series analysis model:

"Prediction of Olympic medals based on time series analysis" uses time series analysis, takes the medals of France in previous Olympic Games as data, and establishes a prediction model in R language to finally predict the medals of France in the next Olympic Games. "Prediction of 2008 Olympic medals using time series method" also uses time series method, based on the medal data of China's 1984-2004 Olympic Games, and predicts the number of Chinese medals in the 2008 Olympic Games by establishing an ARIMA model.

- Regression model:

"Prediction of China's 31st Olympic Games medals based on linear regression dynamic model" constructs an Olympic medal dynamic model, points out the concepts of intrinsic growth rate and maximum capacity, and uses linear regression to predict the medals of China's 31st Olympic Games. By analyzing the results of China's Olympic Games from 1984 to 2012,

the least squares method is used to fit the data and establish a regression equation. "Research on the Prediction of Medal Counts in Large-scale International Comprehensive Sports Games - Taking the Beijing Olympics as an Example" is based on econometric theory, considering factors such as population, per capita GDP, home advantage, social system and the results of the previous Olympic Games, and establishing a preliminary model of multivariate nonlinear regression.

- Grey prediction model and combination model:

"Olympic sports performance prediction model based on sequence and multi-factor analysis" introduces the grey prediction model GM (1,1), accumulates and generates the original data, mines the data rules, and establishes a model. "A new grey Markov prediction model and its application" proposes a grey Markov prediction model, combining the advantages of the grey prediction model and the Markov chain, accumulates and generates the original data and weakens the randomness, uses the Markov chain to predict the state transition, and applies it to the prediction of the number of Olympic medals.

These studies show that different prediction models are suitable for different data characteristics and situations. Time series analysis is suitable for simple, stable and periodic data; the linear regression model can consider the impact of multiple factors on the number of medals; the grey prediction model is suitable for short-term prediction with less data; the combination prediction model can combine the advantages of each single prediction method to improve the accuracy and reliability of the prediction. Future research can further consider more factors that affect the number of Olympic medals, such as politics, culture, and sports policies, and use a variety of prediction methods to build a more complete prediction model.

1.3 Our Work

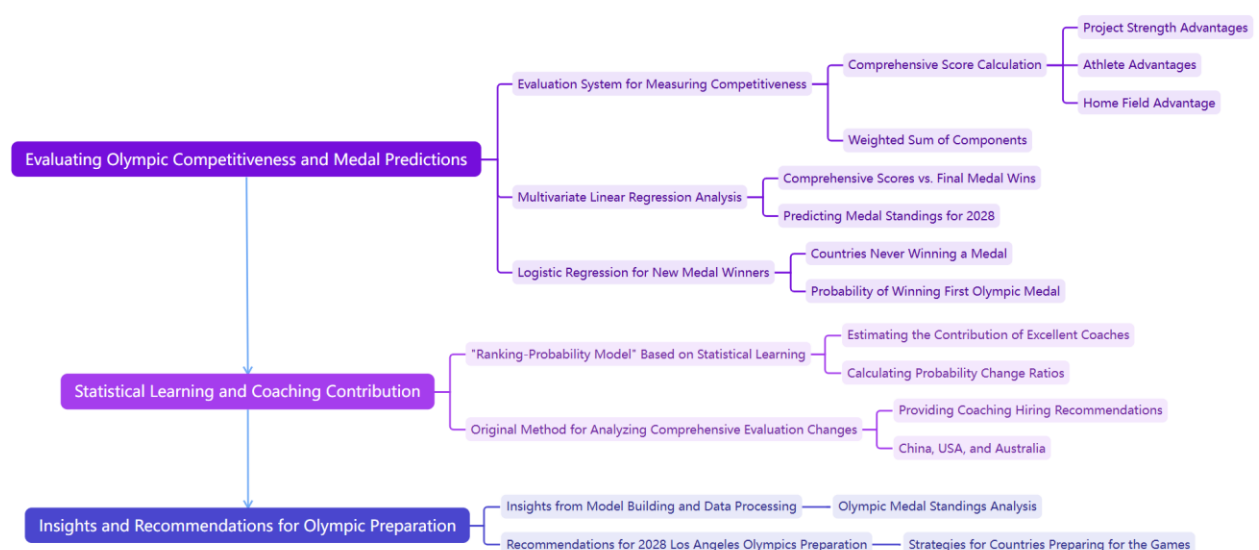


Figure 2 Our work

In order to solve the three problems mentioned above and make appropriate suggestions for the Olympic participating countries, our work is carried out in the following way:

For the first question, we firstly establish an evaluation system to measure the competitiveness of each country in each project in a specific year, which is called the comprehensive evaluation score. The comprehensive evaluation score is obtained by weighting and summing the project strength advantage, athlete advantage, and home advantage according to a certain proportion.

Secondly, use multivariate linear regression to fit the relationship between the comprehensive score of each project in previous years and the final award situation. Then bring in the latest comprehensive score data of each country and each project to estimate the medal list in 2028

Thirdly, we use the logistic regression method to estimate the probability of a country that has never won an Olympic medal to win the first Olympic medal

For the second question, we proposed a "ranking-probability model" based on statistical learning. Thus, the contribution of excellent coaches is estimated by calculating the proportion of probability changes.

Then, we originally proposed a method based on the analysis of changes in project comprehensive evaluation to provide China, the United States and Australia with suggestions on hiring coaches.

For the third question, we summarize the unique insights on the Olympic medal table during the model building and data processing process, and make suggestions to countries in their preparations for the 2028 Los Angeles Olympics.

2 Assumptions and Justifications

1. *In the "ranking-probability model", the probability of winning a medal for a specific ranking will be different due to the difference in strength between different events. But we assume that after we count the results of hundreds of games, the probability of winning a medal for each ranking will converge to a fixed value:*

According to the law of large numbers, the superposition of multiple independent random variables obeys the normal distribution, and the effects of extremely large gaps and extremely small gaps should always be able to offset each other, so that the probability converges.

2. *Assume that for countries that have never won a prize, the host effect has little effect on whether they can win a medal:*

The host effect will affect the number of medals of the host to a certain extent, but it will not make the country that originally has the strength to win a medal unable to get a medal. Therefore, when analyzing the probability of winning a medal for a country that has never won a prize, the impact of the host effect is extremely small.

3. *Assuming that a sport has fewer finals (less than or equal to 5), its contribution to the*

total number of medals is less and cannot be included in the events that are important to the country:

Sports with fewer finals can win fewer medals, so for each country, the important sports are mostly those with more finals.

4. Assuming that great coaches can weaken the negative impact of the host effect, the ability value with a smaller host effect parameter is used when considering the impact of great coaches:

For non-host countries, the great coach effect may lead the players to a better state of play and resist the negative impact of the host effect.

5. Assuming that a country has always been weak in a certain project, the effect of hiring a great coach is not ideal:

The great coach effect only plays a supporting role in the strength of the project. For projects with weak strength, even if there are great coaches, it often cannot achieve obvious results in the short term.

6. Assuming that a country has always been strong in a certain project, the improvement effect after hiring a great coach will not be too obvious:

If a country has always been strong in a certain project, it has a greater chance of winning a medal regardless of whether it has a great coach or not.

7. Assume that the results of each event are independent of each other:

There is empirical reason to believe that the correlation between each major category of events is very small and can be ignored.

3 Notations and Definitions

3.1 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description	Unit
P_i	Project Strength Advantages	\
A_i	Athlete Advantage	\
H_i	Home advantage	\
$Score_i$	Comprehensive evaluation score	\
a_{cast}	Data after mapping to the [0,1] interval	\
θ_i	regression coefficient	\
ϵ_t	error term	\
$gold_{eff}$	The impact of host effect on the number of gold medals	\
$total_{eff}$	The impact of host effect on total medal count	\
$true_{gold}$	The host country's recent actual gold medal level	\
$true_{medal}$	Host country's recent actual total medal level	\

δ_g	Gold medal-contribution ratio	\
δ_s	Silver medal-contribution ratio	\
δ_b	Bronze medal-contribution ratio	\
P_{before}	The probability of obtaining the corresponding results before the arrival of the excellent coach	\
P_{after}	The probability of obtaining the corresponding results after the arrival of the excellent coach	\

3.2 Definitions

- *Comprehensive evaluation score*: Quantified scores of competitiveness of each country and project over the years.
- *Project Strength Advantages*: Comprehensive quantification of factors such as a country's good youth training system, excellent technical and tactical accumulation, and good mass base in a certain project.
- *Athlete Advantage*: A comprehensive quantification of factors such as whether a country has extremely outstanding athletes in a certain event or athletes with Olympic experience.
- *Home field advantage*: Comprehensive quantification of factors such as audience support, logistics support, and venue adaptation for countries playing at home.

4 Model 1: PAH Comprehensive Evaluation Model

4.1 Problem Description

We need to make predictions for the 2028 Los Angeles Olympics.

First, we want to give a forecast of the medal table, including prediction intervals for all results, an estimate of the model's accuracy, and a performance measure. Next, we want to see which countries are likely to improve in the medal table and which countries will perform worse than in 2024. Furthermore, we also want to discover new names in the medal table - countries that have never won an Olympic medal. In this process, we want to consider the importance of different sports to different countries. Finally, we are also interested in the impact of the host country's choice of sports on the medal table.

4.2 PAH Comprehensive Evaluation Model

4.2.1 Calculate the overall evaluation score

We use the "PAH comprehensive evaluation" model to solve the problem of medal table prediction. The key to the model is to give each country a "comprehensive score" in each event every year to quantify the country's competitiveness in the event. Considering the factors that affect this competitiveness:

1. Project strength advantage: that is, a country may have a good youth training system,

excellent technical and tactical accumulation and a good mass base in a certain event. These factors are long-term and stable. Therefore, we can effectively estimate the effect of these factors by analyzing past results.

2. Athlete advantage: that is, a country may have extremely outstanding athletes or athletes with Olympic experience in a certain event, and the personal abilities and experience of these athletes may make important contributions.

3. Home advantage: Countries playing at home often have home advantages. This is reflected in more domestic spectators cheering for them, a good competition environment and logistical support, etc.

Considering the independence of these three advantages, we use:

$$Socre_i = C_1 * P_i + C_2 * A_i + C_3 * H_i \quad (1)$$

to estimate the comprehensive score of a country in a certain event. Among them:

P_i represents Project Strength Advantages, A_i represents Athlete Advantage, H_i represents Home advantage.

Considering that the importance of the three factors may be different when analyzing gold medals and medals, for example, project strength may be more critical in the competition for gold medals, while athlete advantages can often help the host country win more medals. We use the hierarchical analysis method to obtain different parameters. The specific process is as follows:

1. Through the importance of each influencing factor, an importance judgment matrix is constructed. The element in the i-th row and j-th column of the matrix represents the importance of the i-th factor relative to the j-th element.

$$\begin{bmatrix} 1 & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & 1 \end{bmatrix}$$

2. Consistency check: Calculate the maximum eigenvalue V of the matrix, compare it with the order of the matrix, calculate CI and $RI[n]$, if it is less than, the consistency check is passed, otherwise reconstruct the importance judgment matrix. Among them:

$$CI = \frac{V - n}{n - 1} \quad (2)$$

$$RI[n] = [0, 0.0001, 0.52, 0.89, 1.12, 1.26, 1.36, 1.41, 1.46, 1.49, 1.52, 1.54, 1.56, 1.58, 1.59] \quad (3)$$

3. Calculate weight: Find the eigenvector β corresponding to the maximum eigenvalue V of the judgment matrix after the consistency test. Each element of β is the weight of each influencing factor.

$$\beta = \begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix} \quad (4)$$

The calculation results are as follows:

For gold medal: $C_1 = 0.35$ $C_2 = 0.10$ $C_3 = 0.55$

For medal: $C_1 = 0.3325$ $C_2 = 0.1396$ $C_3 = 0.5279$

Now that we have three parameters, let's consider how to calculate the three indicators of project strength advantage, athlete advantage, and home advantage.

- In terms of project strength advantage, we consider all the winning records of the past eight Olympic Games since the 1996 Atlanta Olympic Games.

For a certain event, suppose there are n_j gold medals in the past j Olympic Games, and consider that the weighted total number of medals won by country i in the past j Olympic Games (gold medals * 0.5 + silver medals * 0.3 + bronze medals * 0.2) is a_{ij} , then the country's event strength advantage score is:

$$P_i = \frac{1}{8} \sum_{j=1}^8 \frac{a_{ij}}{n_i} * 1000 \quad (5)$$

- In terms of athlete advantages, we considered athletes who have participated in 1-2 Olympic Games. We believe that these young athletes with certain experience are likely to participate in the Los Angeles Olympics and win glory for the country.

For a certain event, let the number of people from country i participating in the event for the first time in 2020 be a_i , and the number of people participating for the first time in 2024 be b_i , then the score of the country in the athlete advantage item is:

$$A_i = \frac{a_i + b_i}{\sum_i (a_i + b_i)} * 100 \quad (6)$$

- In terms of home advantage,

$$\begin{aligned} H_i &= 1 & (i = \text{host country}) \\ H_i &= 0 & (i \neq \text{host country}) \end{aligned} \quad (7)$$

Considering the different magnitudes of the three indicators, we consider using the following method to map the data to the $[0,1]$ interval

$$a_{cast} = \frac{a_{origin-min}}{max-min} \quad (8)$$

Thus, we get the comprehensive evaluation scores of each country and each event over the years.

4.2.2 Multiple linear regression fitting medal table

We perform multivariate linear regression on the scores of various events in previous years

and the final medal table to fit the medal table in 2028. The specific method is as follows:

- Model formula used:

$$y = \theta_0 + \sum_{i=1}^n \theta_i x_i + \epsilon_t \quad (9)$$

Where y is the regression dependent variable, x_i is the regression independent variable, ϵ is the error term, and θ_i is the regression coefficient.

- Considering that the number of gold medals and medals won by a country is related to its corresponding comprehensive score, the y value here is the number of gold medals and medals, and x_i is the gold medal and medal ability values of each event of the country.

The specific operation method of the model is as follows:

1. Calculate the ability value of each country based on the processed data, and count the number of gold medals and medals of each country each year.
2. Use the “fitlm” function in MATLAB to calculate the regression coefficient, and calculate the predicted value and confidence interval.

4.2.3 Accuracy estimation

Use the “fitlm” function in MATLAB to calculate the regression coefficients and obtain the predicted values and confidence intervals.

4.2.4 Estimated situation for non-awarded countries

Since countries that have never won awards often have less data, conventional methods may have large deviations in estimation and there are only two situations for winning or not, so we specifically use the logistic regression method to estimate. The specific process is as follows:

1. Since the host effect of small countries is roughly the same, the ability values of each country and each project should use the parameters with weaker host effects obtained by the previous analytic hierarchy process, that is:

$$C_1 = 0.6870 \quad C_2 = 0.1865 \quad C_3 = 0.1265$$

2. Regression equation:

$$F(x) = \frac{1}{1+e^{-kx+b}} \quad (10)$$

Where x is the fitted dependent variable and $f(x)$ is the probability of winning the output medal.

3. Since any one project is awarded, it is recorded as winning, so the dependent variable is selected as the sum of the ability values of each project in each country.

$$x_i = \sum_{j=1}^k a_{ij} \quad (11)$$

Where i is the country sequence subscript, j is the sports sequence subscript.

4. Process the past data and calculate the regression parameters k and b .

5. Input the 2028 data into the logistic regression equation, and then select the corresponding values of countries that have never won medals in the past, which is the estimated probability of winning medals in 2028.

4.2.5 The importance of different projects in different countries

Since each country has different advantage events, the number of events in different competitions will lead to changes in the final number of medals. Therefore, multiple linear regression is used to fit the changes in the final number of medals in different events, and the regression coefficient represents the contribution of the event to the country's total number of medals.^[1]

Model:

$$y = \theta_0 + \sum_{i=1}^n \theta_i x_i + \epsilon_t \quad (12)$$

Since there are many competition events, different small events under the same sport should be merged and regressed. The specific process is as follows:

1. Sort out the data from previous years and use the multivariate linear regression algorithm to get the regression coefficient for each sport:

I. First, ordinary linear regression. The p -value of the regression result is large, so linear regression should not be used.

II. For China, the logarithm of the total number of medals is taken, and the p -value of the regression result is small. Therefore, the logarithmic linear regression of the total number of medals is used, and the regression model is:

$$\ln(y) = \theta_0 + \sum_{i=1}^n \theta_i x_i + \epsilon_t \quad (13)$$

III. For the United States, the total number of medals is divided by 100 and raised to the 10th power. The regression p -value is small, and the regression model is:

$$e^{\frac{y}{100}} = \theta_0 + \sum_{i=1}^n \theta_i x_i + \epsilon_t \quad (14)$$

IV. For Japan, the logarithm of the total number of medals is taken, and the p -value of the regression result is small. Therefore, the logarithmic linear regression of the total number of medals is used, and the regression model is:

$$\ln(y) = \theta_0 + \sum_{i=1}^n \theta_i x_i + \epsilon_t \quad (15)$$

2. Compare the regression coefficients of different sports in the same country. The one with the larger regression coefficient is the one that is more important to the country.

4.2.6 Host Effect

The main ideas for considering the impact of the events selected by the host on the medal table are as follows:

1. Due to factors such as the degree of adaptation to the competition venue, the audience at the competition venue, and the distribution of competition events, the host may have certain advantages in the Olympics.^{[2][3]}
2.
 - Establish a model to measure the impact of the host effect on the number of gold medals.

$$gold_{eff} = \frac{x_{gold\ of\ host}}{x_{gold\ of\ total}} \quad (16)$$

That is, the number of gold medals of the host is divided by the total number of gold medals in the Olympic Games.

- Establish a model to measure the impact of the host effect on the total number of medals.

$$total_{eff} = \frac{x_{total\ of\ host}}{x_{total\ of\ total}} \quad (17)$$

That is, the total number of medals of the host is divided by the total number of medals of the Olympic Games.

- Establish a model to measure the actual gold medal level of the host in recent times.

$$true_{gold} = \frac{x_{host}}{x_{total}} \quad (18)$$

That is, the total number of gold medals won by the host country in the first two Olympic Games is divided by the total number of gold medals in the previous two Olympic Games

- Establish a model to measure the host country's recent actual total medal level.

$$true_{medals} = \frac{x_{medals\ for\ last\ two}}{x_{medals\ of\ total}} \quad (19)$$

That is, the total number of medals won by the host country in the first two Olympic Games is divided by the total number of medals in the previous two Olympic Games.

Selecting all previous hosts and calculating the above ratios can reflect the host effect.

4.3 Results

4.3.1 Calculate the comprehensive evaluation score

Use the code in the appendix to process the data according to the method mentioned in 4.2.1 to obtain the comprehensive evaluation scores of each country and project over the years. Here is a part of it as an example:

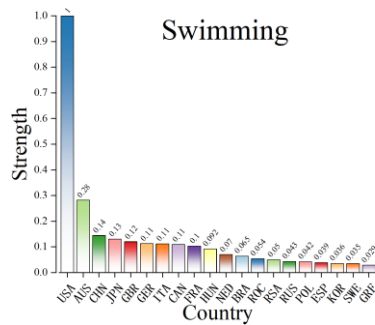


Figure 3 Swimming scores

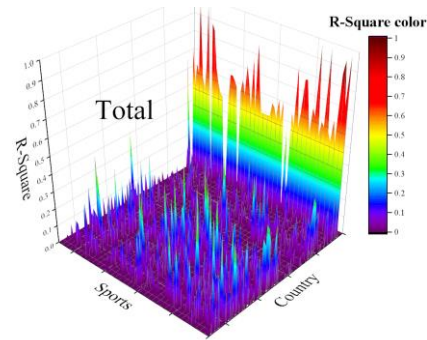


Figure 4 Evaluation score

4.3.2 Fitting the Medal Table Using Multiple Linear Regression

We used the linear regression function of MATLAB to fit the corresponding parameters, and then plugged in the comprehensive evaluation scores of various countries and events in the past years after 2024 to get our final predictions (including confidence intervals) for the medal and gold medal tables in 2028. The top 20 are shown here:

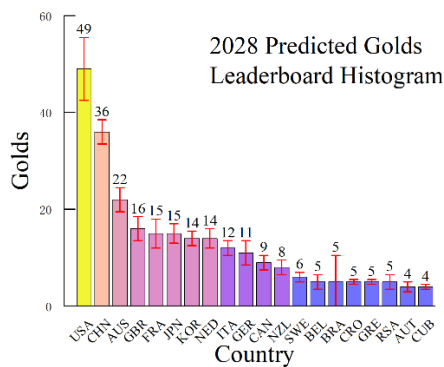


Figure 5 Gold table prediction

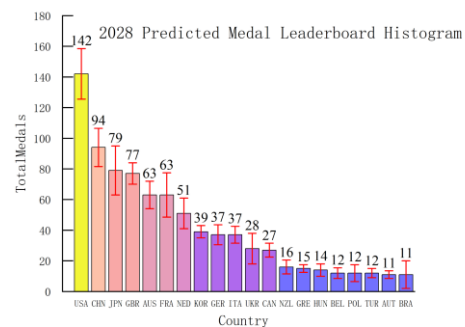


Figure 6 Medals table prediction

4.3.3 Countries that are improving and declining

Comparing the medal table in 2024 with our predicted medal table will reveal the countries that have improved and declined. For the sake of brevity, we will only summarize the progress and decline of the top 20 countries here:

- Countries expected to improve include: the United States, Japan, Australia, the United Kingdom, etc.
- Countries that remain relatively stable include: China, France, Italy, Germany, etc.
- Countries that may decline include: New Zealand, Spain, Uzbekistan, etc.

4.3.4 First-time medal winning countries

Comparing the medal table, we believe that the following countries are likely to win their first Olympic gold medal, and estimate the probability according to the algorithm in 4.2.4:

American Samoa: 0.2071

Angola: 0.2068

Bangladesh: 0.2007

Democratic Republic of the Congo: 0.2004

Morocco: 0.2004

Papua New Guinea: 0.2003

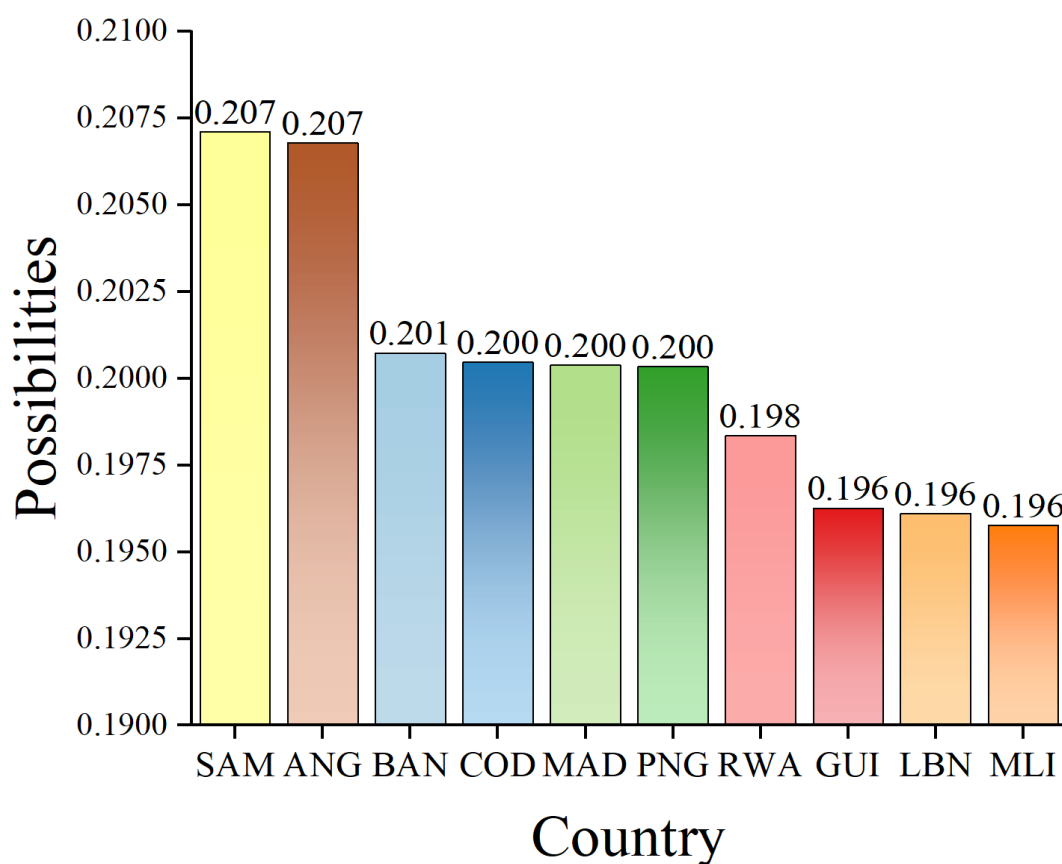


Figure 7 Probability of winning first Olympic medal

4.3.5 The importance of different projects in different countries

We considered using the method mentioned in 4.2.5 to analyze the events that are important to China, the United States, and Japan, and the results we obtained are as follows:

1. For China, the most important events are track and field events and gymnastics events.
2. For the United States, the most important events are gymnastics events.
3. For Japan, the most important events are gymnastics events and judo events.

4.3.6 Host Effect

Our conclusions and suggestions are as follows:

Conclusion:

1. The host effect has a great impact on the gold medals and total medals of the host country. The host effect was particularly significant in the two Olympic Games held in Atlanta and Beijing.

2. The host country is more likely to win more gold medals and medals, which may be due to factors such as the degree of adaptation to the competition venue, the motivation of the audience at the competition venue, and the distribution of competition events.

Suggestions:

1. For countries that are not hosts, they should try their best to create good competition conditions for their own country by paying attention to venue adaptation and doing a good job of logistics support.

2. As a host country, the country can arrange some sports events that suit its own characteristics and make reasonable use of the host effect to make itself more advantageous.

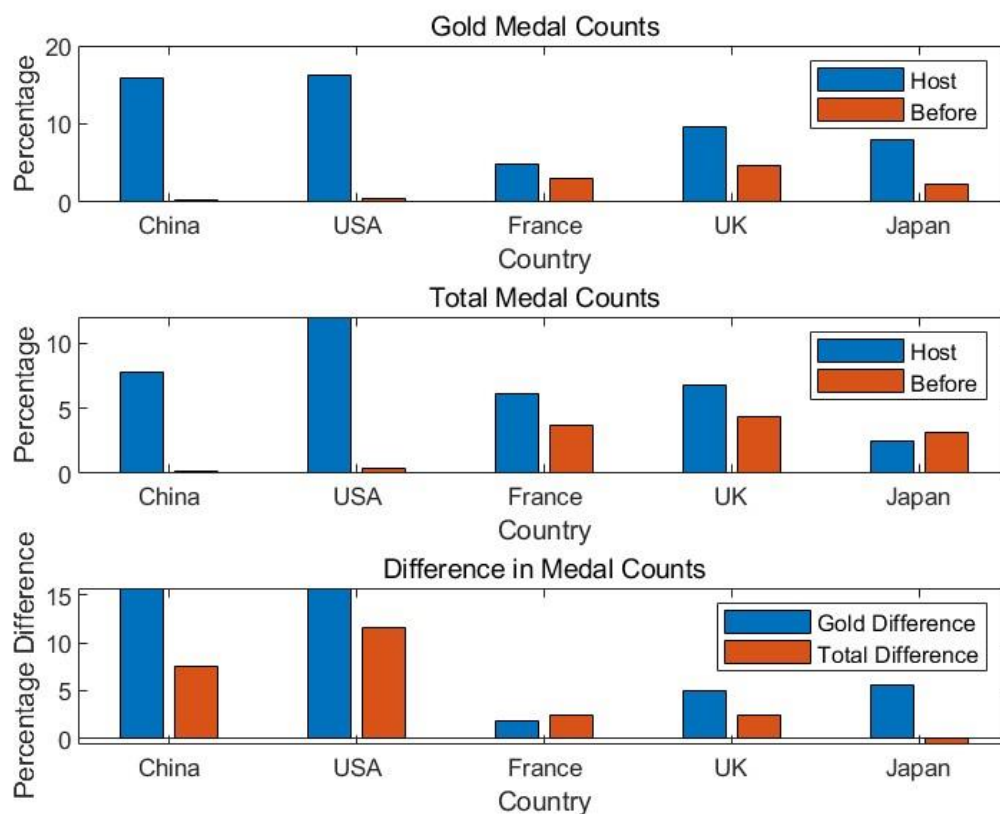


Figure 8 Host effect

5 Model 2: Ranking-Probability Model

5.1 Description of “Great Coach” effect

The "great coach" effect refers to the situation where an excellent coach can effectively guide athletes in training and competition, bringing them obvious progress. The contribution of a great coach to athletes and the country may be reflected in advanced technical and tactical

guidance, helping players to establish a good mentality, and even bringing advanced youth training concepts to the country. Lang Ping, who coached the US and Chinese women's volleyball teams and won championships, and Bella Karolyi, a gymnastics coach who coached the Romanian women's team and the US women's team and achieved great success, are good examples.

We hope to use the "ranking-probability model" to verify the "great change" brought about by the "great coach" and estimate the contribution of this effect to the number of medals, taking the outstanding performance of the Chinese boxing team in Paris in 2024 as an example. Finally, we also hope that the "ranking-probability model" can help us make good recommendations for the United States, China and Australia in hiring coaches.

5.2 Ranking-Probability Model

In order to verify whether a certain excellent result is affected by the "great coach effect", a natural idea is to quantify the probability of achieving the result before the coach comes, P_{before} , and the probability after the coach comes, P_{after} . If P_{after} is significantly greater than P_{before} , then we have reason to believe that the coach's arrival has contributed to the team's excellent results.

To measure the contribution of this "great coach effect" to the number of medals, we define:

Gold medal-contribution ratio:

$$\delta_g = \frac{P_{g_after}}{P_{g_before}} \quad (20)$$

Silver medal-contribution ratio:

$$\delta_s = \frac{P_{s_after}}{P_{s_before}} \quad (21)$$

Bronze medal-contribution ratio:

$$\delta_b = \frac{P_{b_after}}{P_{b_before}} \quad (22)$$

Based on the work of model 1, we quantified the comprehensive strength of various countries in various projects over the years. The next work we have to do is to calculate the probability of the country winning the gold medal, silver medal, and bronze medal under this comprehensive strength. Considering the following facts:

1. Winning a medal is related to the relative competitiveness of the country.
2. The ranking of a country's comprehensive strength can reflect this relative competitiveness.

We decided to establish a "ranking-probability model" to calculate the probability of winning the gold medal, silver medal, and bronze medal for each comprehensive strength ranking. It should be pointed out that although the gap between the various rankings in different projects is different, the probability of winning the same ranking in different projects is not the same.

However, we assume that when the learning data set is large enough, the impact of such large and small differences can always offset each other, so that the probability of winning each place converges in the end. The specific method of using the "ranking-probability model" is as follows:

1. Using all the events of the 2020 and 2024 Olympic Games as training sets, calculate the comprehensive strength of each event in each country before the 2020 Tokyo Olympics and the strength of each event in each country before the 2024 Paris Olympics.
2. For each event, count the frequency of countries ranked 1-12 winning various medals.
3. When the sample is large enough, use the frequency to estimate the probability and get the "ranking-probability table" corresponding to the probability of each medal.
4. Using the "ranking-probability table", calculate the probability of obtaining the corresponding results before and after the coaching.
5. When the ratio of the probability after coaching to the probability before coaching is greater than a certain value, we believe that the coach's arrival has made a significant contribution.

Finally, we also hope to use our work to make recommendations for the three countries in hiring excellent coaches, taking into account the following basic facts:

- Since great coaches do not affect the advantages of the host country, the parameters obtained by the previous part of the hierarchical analysis method can be used here to calculate their ability values.
- Hiring great coaches may require a certain amount of economic strength.
- For a country, if the strength of the project itself has always been weak, the effect of hiring a great coach may still be less than ideal.
- For a country, if the ability value of the project itself has always remained at a high level, the improvement after hiring a great coach may not be obvious.

For the first consideration, we use the parameters obtained by the previous part of the hierarchical analysis method to calculate its ability value, namely:

$$C_1=0.6870, C_2=0.1865, C_3=0.1265 \quad (23)$$

For the second consideration, we decided to select China, the United States, and Australia for analysis.

For the third consideration, based on our analysis and testing, we decided to select projects with ability values greater than 0.3 for analysis.

For the fourth consideration, we decided to select projects whose capacity values have been declining in the past three years for analysis.

The specific process is: use the evaluation model in the first question to calculate the capacity of each project in these three countries, and determine whether the capacity values of each project have been decreasing in the past three years and are all greater than 0.3.

5.3 Results

Given that the Chinese boxing team performed outstandingly in the 2024 Paris Olympics,

winning three golds and two silvers, we considered testing whether the "excellent coach effect" made a significant contribution to this outstanding performance.

After that, we calculated the comprehensive scores of each project in the United States, China, and Australia over the years, and selected projects with scores above 0.3 and a downward trend. We believe that these projects need to hire great coaches.

5.3.1 Parameter Estimation

We have calculated the correspondence between the final medal winners and pre-match rankings for the Tokyo and Paris Olympics, and the results are as follows:

Table 2: Ranking-Probability Table

Ranking	Gold Rate	Silver Rate	Bronze Rate
1	34.04%	26.63%	10.27%
2	18.07%	16.75%	6.16%
3	16.26%	6.98%	8.22%
4	7.83%	8.67%	8.90%
5	4.82%	7.11%	11.64%
6	2.11%	1.33%	9.59%
7	2.71%	3.98%	7.73%
8	3.31%	2.77%	7.51%
9	1.20%	4.10%	3.42%
10	0.90%	2.41%	4.59%
11	1.22%	2.65%	3.44%
12	0.61%	1.08%	1.36%
Others	6.93%	15.54%	17.12%

5.3.2 Calculation Results

According to our evaluation model, Chinese boxing ranks seventh after the Tokyo Olympics, and their probability of winning three golds and two silvers is:

$$P_{before} = 0.0211^3 * 0.0133^2 = 1.662 * 10^{-6} \quad (24)$$

After the Paris Olympics, China's boxing ranking rose to fourth place. Their probability of winning three gold medals and two silver medals is:

$$P_{after} = 0.0783^3 * 0.0867^2 = 3.608 * 10^{-6} \quad (25)$$

$$\alpha = \frac{P_{after}}{P_{before}} = 2.171 \quad (26)$$

Considering that the probability increase is more than 100%, we have reason to believe that the "great coach" has made a significant contribution to the improvement of Chinese boxing performance.

The following calculates the coach's gold medal-contribution ratio, silver medal-contribution ratio, and bronze medal-contribution ratio to quantify this contribution:

$$\text{Gold medal-contribution ratio: } \frac{0.0783}{0.0271} = 2.889 \quad (27)$$

$$\text{Silver medal-contribution ratio: } \frac{0.0867}{0.0398} = 2.178 \quad (28)$$

$$\text{Bronze medal-contribution ratio: } \frac{0.0890}{0.0773} = 1.151 \quad (29)$$

The following considers the changes in the comprehensive strength of various projects in the United States, China, and Australia over the years, and here are our suggestions:

1. For China, excellent coaches should be hired in volleyball and fencing.
2. For the United States, excellent coaches should be hired in Sailing.
3. For Australia, excellent coaches should be hired in Rowing.

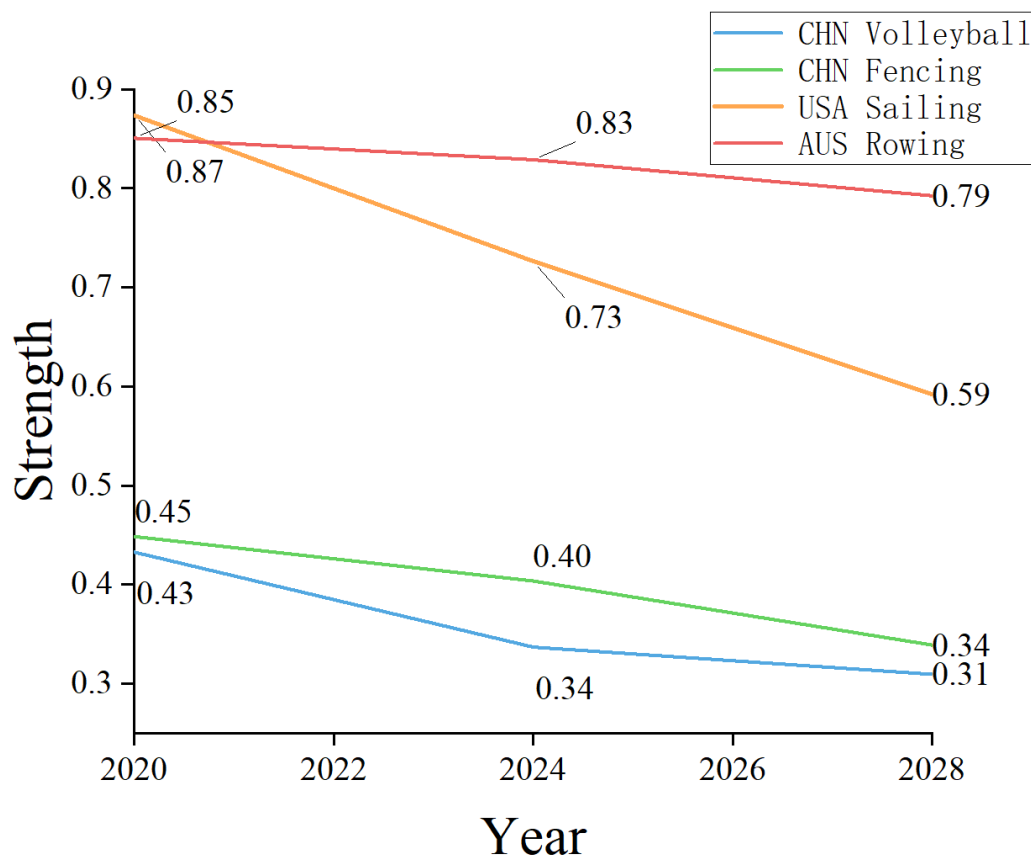


Figure 9 Rating changes of some projects in some countries

6 Insights and Suggestions

6.1 Insights

1. From our "ranking-probability model", we can see that the top three countries in the comprehensive evaluation score eventually won 68.37% of the gold medals, 50.36% of the silver medals, and 24.65% of the bronze medals. This shows that the traditional "powerful countries" have a strong monopoly on gold medals. For countries that have never won an award

in this event, they are relatively more likely to compete for a bronze medal.

2. Also from the "ranking-probability model", we can see that compared with gold and silver medals, the competition for bronze medals is more intense, which is reflected in the fact that the probability of countries ranked first to eighth winning bronze medals is very close.

6.2 Suggestions

1. From the analysis of the host effect, we have witnessed that the host may have a better performance in the competition. We believe that this may be due to factors such as the motivation of the home audience, sufficient adaptation to the venue, and perfect logistical support. Therefore, we also recommend that all participating countries do a good job of security, all athletes fully adapt to the venue before the game, and encourage sports fans to come to the scene to cheer for their country's representative team.

2. From the analysis of hiring excellent coaches, it can be seen that each country should pay attention to those projects with a certain foundation and whose comprehensive evaluation scores are declining year by year.

7 Sensitivity Analysis

7.1 Model stability analysis

With a step size of 0.05, ensuring that $c_1 + c_2 + c_3 = 1$, traverse c_1 , c_2 , c_3 , with a step size of 4, traverse from 4 to 12 years, and obtain the following figure:

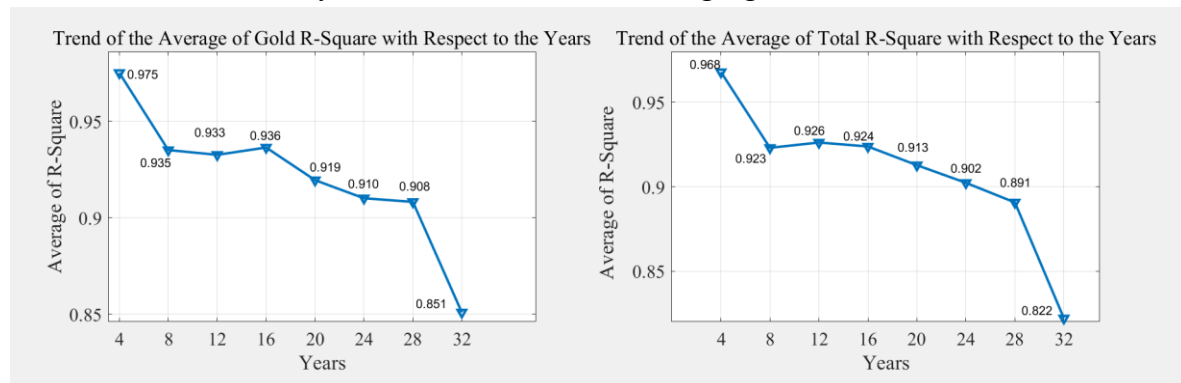


Figure 10 Schematic diagram of the fitting effect changing with the year

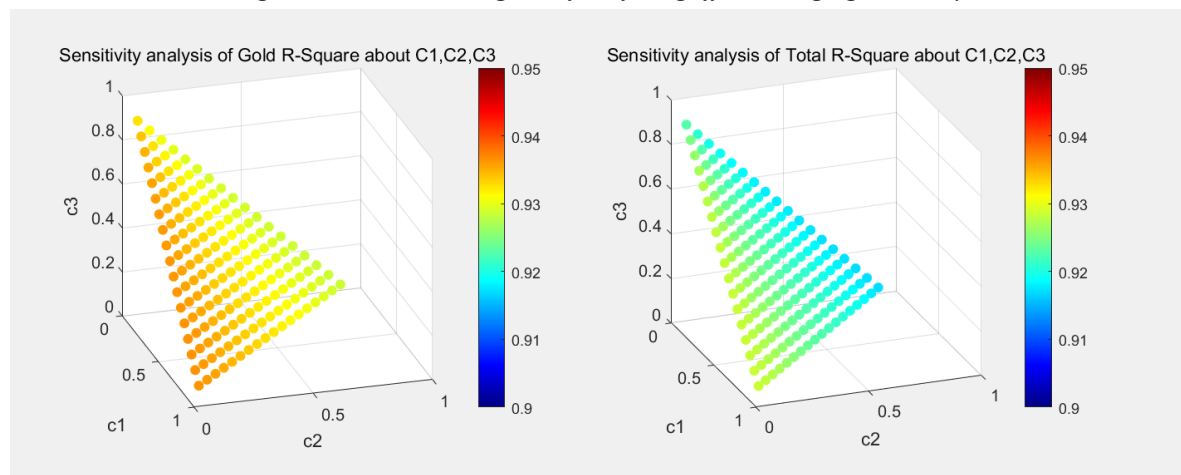


Figure 11 Schematic diagram of parameter changes over time

It can be seen that with the changes of the four parameters, R-square basically remains above 0.9, which shows that the model effect is very stable and the appropriate parameters can be selected according to the actual situation.

7.2 Specific analysis of sensitivity data sets

For C_1 , C_2 and C_3 , it can be seen that when C_1 is particularly large, the R-square is the largest. This is because the national strength is calculated by the proportion of medals and has a strong linear correlation with the number of medals. When C_1 remains unchanged, the larger C_2 is, the smaller the R-square is. C_3 shows that the R-square at both ends is small.

For the year, it can be seen that as the number of years considered for forward fitting increases, the R-square drops sharply, stabilizes, and then drops sharply. It is not difficult to imagine that if only the data of the previous year is used for fitting, there may be a false good fitting effect caused by too little data, and too many years may cause too much noise interference, and the best one after excluding 4 years is 8 or 12 years

Therefore, C_3 should be taken as the middle value of about 0.5, C_2 should be small, C_1 should be appropriate, and the year should be selected as 8 or 12 years as much as possible.

8 Model Evaluation and Further Discussion

8.1 Strengths

1. Strong scalability: The evaluation of ability values can be used to predict medals, guide the selection of great coaches and other aspects. A complete set of ability value judgments can also be used to guide various countries or teams to improve their own strengths, recognize their own shortcomings and advantages, and make more scientific decisions.

2. Low data dependence: All model work only uses the medal data of previous years, the participation of athletes, and the arrangement of events. No other additional data is required, so that the model does not have to rely on other large amounts of data. It can play its role well when facing a large amount of data that is difficult to count.

3. Convenient parameter adjustment: You only need to modify the coefficients to achieve the calculation of different weight considerations. For different problems, you only need to reconstruct the coefficients according to the relative importance of each influencing factor to solve different problems without having to start building the model again.

8.2 Weaknesses

1. Since a linear regression model is used to predict the number of medals, it may be affected by some extreme cases, resulting in poor fitting results. At the same time, whether the ability values of each item are completely independent, if they are not completely independent, it is not appropriate to use the multivariate linear regression method.

8.3 Further Discussion

1. Consider polynomial fitting or exponential logarithmic fitting. Since the correspondence between the number of medals and the ability value of each project is not necessarily completely linear, polynomial fitting or exponential logarithmic fitting can be used to improve

the fitting effect of the model.

2. The multi-model prediction method can be used to change the various parameters involved in the fitting process, such as the three parameters that determine the ability value, and use multiple models to predict separately. For the weights between multiple models, the medal data of previous years and the predicted data of each model can be used. Heuristic algorithms such as simulated annealing can be used for planning, so that the gap between the predicted value and the actual value is minimized, so as to better predict the medal list.^[4]

3. The results of other projects besides the Olympics can be considered to better evaluate the project ability value, such as: World Championships, etc.

9 Conclusion

The main conclusions and contributions of this paper are summarized as follows:

1. After comparative analysis, it is found that countries such as the United States and Japan are expected to improve their medal performance in future Olympic Games. The reason is that these countries have shown a positive development trend in the accumulation of project strength, athlete training and reserve; while New Zealand, Spain, Uzbekistan and other countries may face the challenge of declining medal numbers.
2. American Samoa, Angola and other countries are expected to win their first Olympic medals in the Los Angeles Olympics.
3. In addition, the important position of different projects in the sports strategies of various countries is deeply analyzed. For example, China's track and field and gymnastics, the United States and Japan's gymnastics and other projects have made outstanding contributions to the total number of medals in their own countries, which provides a key basis for countries to reasonably allocate sports resources and formulate targeted training plans.
4. Research on the host country effect shows that the host country does have significant advantages in the Olympic Games. Based on the analysis of the advantages of the host country, we proposed coping strategies such as optimizing competition conditions and strengthening logistics support for non-host countries, and provided suggestions for the host country to reasonably arrange events and give full play to the home advantage.
5. With the help of the ranking probability model, we provide targeted suggestions for the United States, China and Australia on coach hiring decisions. For example, China's volleyball and fencing, the United States' sailing, and Australia's rowing can focus on introducing excellent coaching resources to help project development.

10 References

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Appendices

Appendix 1

Introduce: Project strength score calculation code

```
for i = 1:length(unique_sports)
    for k = 1:length(unique_countries)
        avgratiosum_gold = 0;
        avgratiosum_total = 0;
        for t = 1:length(recent_years)
            avgratiosum_gold = avgratiosum_gold + gold_medal_year_country(t, k);
            avgratiosum_total = avgratiosum_total + total_medal_year_country(t, k);
        end
        country_data{k, 1} = unique_countries{k};
        country_data{k, 2} = 1000 * avgratiosum_gold / length(recent_years);
        country_data{k, 3} = 1000 * avgratiosum_total / length(recent_years);
        country_data{k, 3} = fillmissing(country_data{k, 3}, 'constant', 0);
        country_data{k, 2} = fillmissing(country_data{k, 2}, 'constant', 0);
    end
end
```

Appendix 2

Introduce: Comprehensive score calculation code

```
for i = 1:length(unique_sports)
    sport = unique_sports{i};
    fileID = fopen(fullfile(output_folder, [sport '.txt']), 'w');
    sport_filter = strcmp(sports, sport) & athletes_before2;
    unique_countries = unique(countries(sport_filter));
    for j = 1:length(unique_countries)
        country = unique_countries{j};
        country_filter = strcmp(countries, country) & sport_filter;
        athlete_count = length(unique(names(sport_filter)));
        only_before2_filter = country_filter & ismember(years, [current_year - 8, current_year - 4]);
        unique_athletes_before2 = unique(names(only_before2_filter));
        only_before2_count = length(unique_athletes_before2);
        ratio = only_before2_count / athlete_count;
    end
    fclose(fileID);
end
```